

ArcEscrow

Agentic Letter of Credit for Cross-Border SME Trade on Arc

Submission Document - Stablecoins Commerce Stack Challenge

This document summarizes the project title, description, selected track, Circle product usage, working MVP, architecture, video demo, documentation, and product feedback for upload with the final submission.

Project title	ArcEscrow - Agentic Letter of Credit
Track	Best Agentic Economy Experience on Arc
Submission status	Ready for initial submission as a functional Arc Testnet MVP; strongest remaining evidence upgrade is a complete wallet-signed ERC-8183 tx sequence.
Circle account email	Owner-provided at final submission time. This value is intentionally not stored in the public repository.
Live demo	https://arc-agentic-settlement-lab.vercel.app
Demo video	https://arc-agentic-settlement-lab.vercel.app/demo/arc-escrow-demo.mp4
GitHub repository	https://github.com/a941249849/arc-agentic-settlement-lab

1. Project Description

ArcEscrow is a programmatic escrow product for cross-border SME trade on Arc Testnet. It turns a supplier payment into a proof-gated commercial settlement: USDC is locked in an ERC-8183 lifecycle, agent identity is checked through ERC-8004, delivery evidence is reviewed, and the final receipt binds trade context with transaction evidence.

The product is intentionally not a generic wallet-transfer demo. It models a buyer, supplier, and evaluator workflow similar to a programmable letter of credit for small business trade settlement.

2. Products And Primitives Used

Product / Primitive	Status	Use in ArcEscrow
USDC on Arc	Implemented	Settlement asset and gas-denominated Arc Testnet rail.
Arc Testnet	Implemented	Execution environment for identity reads and ERC-8183 wallet transaction preparation.
ERC-8004	Implemented	IdentityRegistry ownerOf and tokenURI reads for agent proof.
ERC-8183	Implemented	AgenticCommerce job lifecycle: createJob, setBudget, approve, fund, submit, complete.
Circle Wallets	Next integration	Policy-controlled agent treasury and non-crypto-native onboarding.

Product / Primitive	Status	Use in ArcEscrow
Gateway / Nanopayments	Next integration	Paid access layer for reports, document checks, and agent-to-service calls.
CCTP / Bridge Kit	Optional next step	Buyer funding path when capital originates outside Arc.
USYC / StableFX	Gated / conceptual	Not claimed as live; potential treasury or FX extensions if access is granted.

3. Working MVP

- Public Next.js application deployed on Vercel with overview, deal room, agent proof directory, and submission evidence pack.
- Deal room for creating trade settlement requests with buyer, supplier, evaluator, amount, invoice, route, compliance, and deliverable context.
- ERC-8004 identity proof page that prepares registration calldata and verifies existing agent identity by reading Arc Testnet contracts.
- ERC-8183 wallet-execution control path for createJob, setBudget, approve, fund, submit, and complete actions.
- Evaluator gate that reviews trade context, budget, proof, identity, and tx evidence before release.
- Deterministic JSON and Markdown receipt export after Arc transaction evidence is attached.
- Hosted 45-second video walkthrough and public documentation package.

4. Architecture Diagram

Business user -> Buyer agent -> Budget policy -> ERC-8004 identity check -> ERC-8183 createJob -> Provider setBudget -> USDC approve/fund -> Deliverable hash -> Evaluator approval -> Settlement receipt

Circle Wallets and Gateway/Nanopayments are documented as the next integration layer: Wallets for policy-controlled agent treasury and Gateway/Nanopayments for paid API/document workflows. CCTP, USYC, and StableFX are treated as optional or gated extensions, not live MVP claims.

5. Video Demo And Documentation

Video demo	https://arc-agentic-settlement-lab.vercel.app/demo/arc-escrow-demo.mp4
Submission pack page	https://arc-agentic-settlement-lab.vercel.app/challenge
Deal room	https://arc-agentic-settlement-lab.vercel.app/jobs
Agent proof page	https://arc-agentic-settlement-lab.vercel.app/identity
GitHub README	https://github.com/a941249849/arc-agentic-settlement-lab
Arcscan	https://testnet.arcscan.app

6. Product Feedback For Circle / Arc

- A standard agentic commerce reference app combining Circle Wallets, Gateway/Nanopayments, ERC-8004, and ERC-8183 would materially reduce builder uncertainty.
- Arc Testnet would benefit from deterministic sample jobs, funded test identities, faucet status, and reproducible tx hash examples for reviewer-facing demos.
- Stablecoin commerce workflows need a standard settlement receipt schema that binds business context, agent identity, deliverable proof, lifecycle state, and tx hashes.
- Documentation should clearly separate public testnet-ready products from gated, enterprise-only, or conceptual primitives during public builder programs.

7. Submission Honesty Gate

Claim as live: USDC on Arc, Arc Testnet app, ERC-8004 identity reads, ERC-8183 transaction preparation / wallet execution controls, deal lifecycle, evaluator review, and deterministic receipt export.

Do not claim as live unless separately completed: Circle Wallets, Gateway/Nanopayments, CCTP / Bridge Kit, USYC, StableFX, or a full onchain-verified ERC-8183 sequence. The current strongest evidence upgrade is to run createJob -> setBudget -> approve -> fund -> submit -> complete with wallet-signed transaction hashes and attach those hashes to the exported receipt.

8. Short Submission Description

ArcEscrow is an Agentic Letter of Credit on Arc: an SME trade settlement product where importer agents settle cross-border invoices with USDC, verify agent identity, enforce supplier budgets, escrow settlement, bind deliverable proof, and export auditable receipts. It targets the overlap between cross-border stablecoin payments, SME trade workflows, and the agentic economy.